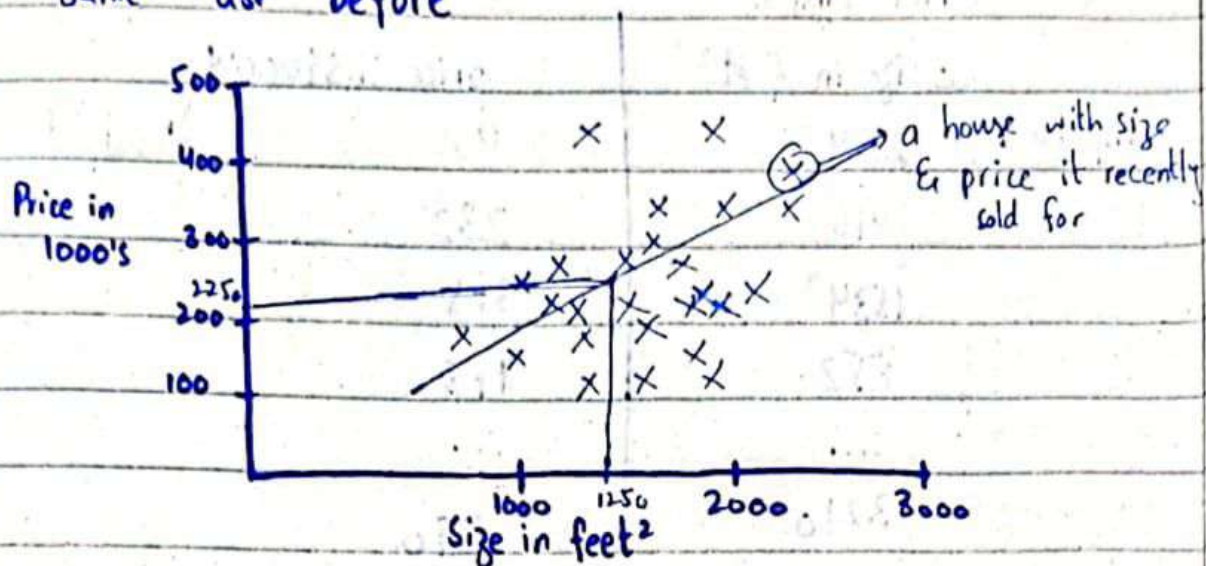


Linear Regression with one variable

Predicting price of house based on their sizes.
same as before



a house is 1250 sq. feet. you wanna sell...
→ one thing you can do is build a linear regression model. acc to above diagram, the house will be 225k Million dollars.

→ Regression model → predict numbers (225k, 1.5, -33.2).

→ Linear Regression is a type of (example of) regression model. • infinite possible outputs

in contrast

→ Classification model → predict categories (if pic is cat or not) or a disease or not
• less possible outputs.

• discrete finite set of possible outputs.

→ In addition to visualizing data as graph plot (as before) there's another way of looking at data that could be useful. (data table).

inputs	Size in feet ²	price in \$1000's	outputs
	2104	400	
	1416	232	
	1534	315	
	852	178	
	...	...	
	3210	870	

- if there 37 rows in data table, there would be 37 crosses each ^{cross} row corresponds to one row of table.

Terminology:

- **Training set:** Data used to train the model.

- x : "input" variable (also feature) e.g. $x = 2104$
(Size in feet²)

- y : "output" variable, or "target" variable

$$y = 400$$

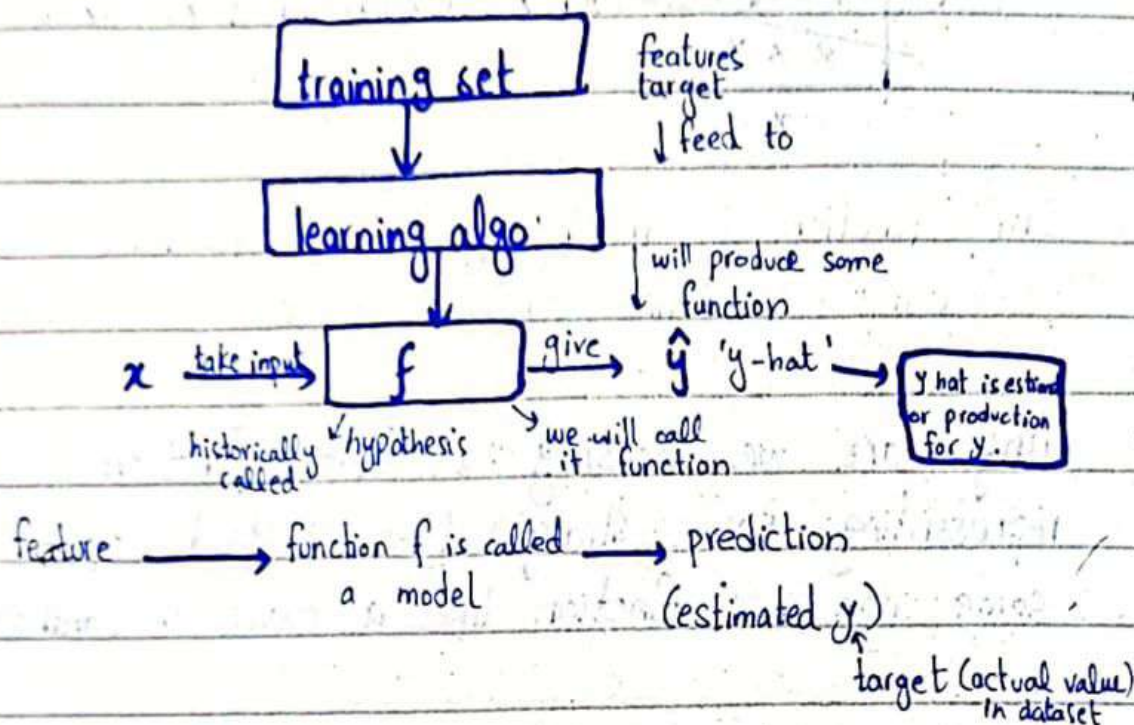
- m : no. of training examples.

$$m = 37$$

- (x, y) : single training example $(2104, 400)$

$(x^{(i)}, y^{(i)})$: refer to specific training example
x subscript i (specific row in this table to left).

y i i^{th} training example.
 $(x^{(1)}, y^{(1)}) = (2104, 400)$



→ if you are helping client sell a house, true price of house is unknown until they sell it.
size → f → price(estimated)

Now,

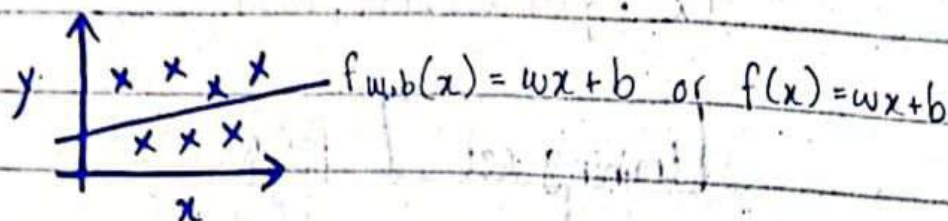
when we are designing an algorithm: True y is:
How to represent f ?

what's the math formula we gonna use to compute f . (for now stick with f being a straight line). function can be written as:

$$f_{w,b}(x) = wx + b$$

(w & b will discussed later, for now they are numbers that on which prediction $\hat{y}$ will be based on)

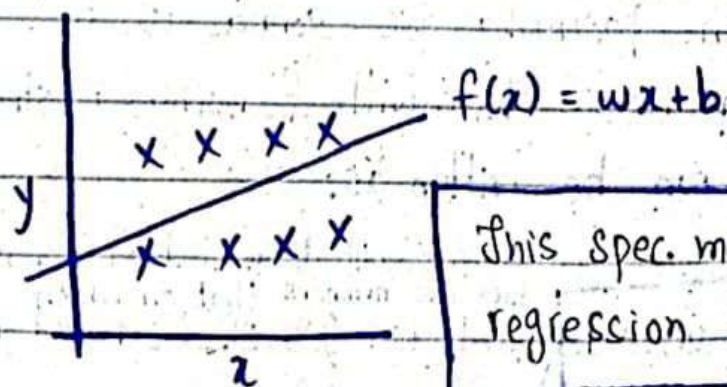
So, f is a function that will take input (x) , and depending on values of b & w it will give a prediction $\hat{y}$.



The function is making predictions for value of y using streamline function of x .

Why are we choosing a linear function representing some straight line instead of some non-linear function like a curve or parabola?

Sometimes you wanna fit more complex non linear functions as well, like a curve but a linear function is simple and easy to work with. Let's use this as a foundation that will eventually help you to get to more complex non-linear models.



This spec. model is called linear regression with one variable.

one variable means single input or feature x .
→ another name for this is

“univariate linear regression”
one variable